

BTech→Big Tech

My Journey, GPU Research and the Mindset for AI world



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10-Oct-2026

Outline

- My Journey to IITM
 - Undergrad
 - Admissions
 - Research/Life at IITM
- GPU Research
 - History of GPGPU
 - Parallel Programming languages
 - Tools, Libraries, Profilers
- Industry & AI Mindset
 - Scenario right now
 - Semi-Conductor Industry
 - General Guidelines

Motivation

- To encourage students to join Research institutions such as IITs/..
- Research exposure gives headstart to your career

Seeking to find answers to the unknown (why?)
Questions are a natural thing of
a curious mind.



My Journey

- BTech in TCE. Worked in IT companies.
- GATE Prep from Standard Books
- Some courses NPTEL Lectures. Eg. TOC

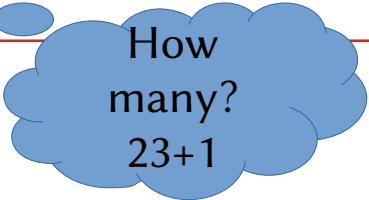
- Masters **Theme:** Graph algorithms
- PhD **Theme:** Accelerating Graph Algorithms on GPUs.

- Now, online BS degree programs with multiple exit options.

Admission to IITx/IISc

- M.Tech

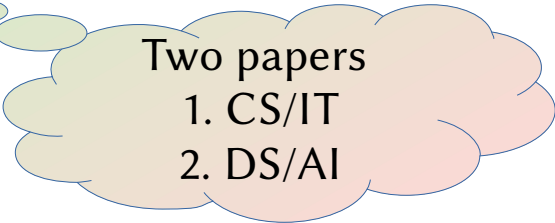
- GATE Exam <https://gate.iitm.ac.in>
- AIR ~150-1100 <http://mtechadm.iitm.ac.in>
- Common Offer Acceptance Portal (COAP for IIT and CCMT for NIT)
- Based only GATE scores
- GATE Rank + Interviews //Some IIT M.Techs



How
many?
23+1

- Direct PhD and M.S (Research)

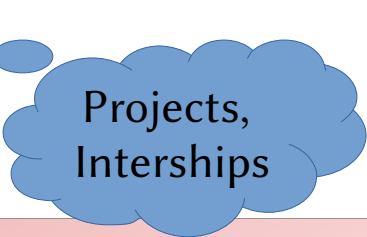
- AIR ~200-2000
- Written test + Interview <https://research.iitm.ac.in>



Two papers
1. CS/IT
2. DS/AI

- PhD

- Masters + Bachelors CGPA
- Written test (GATE Level)
- Interview to two areas.



Projects,
Internships

CS Research

- Theory CS
- Systems
- Intelligent Systems

CS Research

- **Theory CS**
 - Algorithms, Graph theory,
 - Logic, Complexity,
 - Cryptography, ..
- Systems
- Intelligent Systems



Thanks to <https://www.csa.iisc.ac.in/category/research/>

CS Research

- Theory CS
- **Systems**
 - Compilers, Networks
 - DBMS, Architecture,
 - OS, Wireless,
 - Embedded, **Parallelization**
- Intelligent Systems



Thanks to <https://www.csa.iisc.ac.in/category/research/>

CS Research

- Theory CS
- Systems
- **Intelligent Systems**
 - AI, ML, DL,
 - NLP, PR
 - Game theory, DS

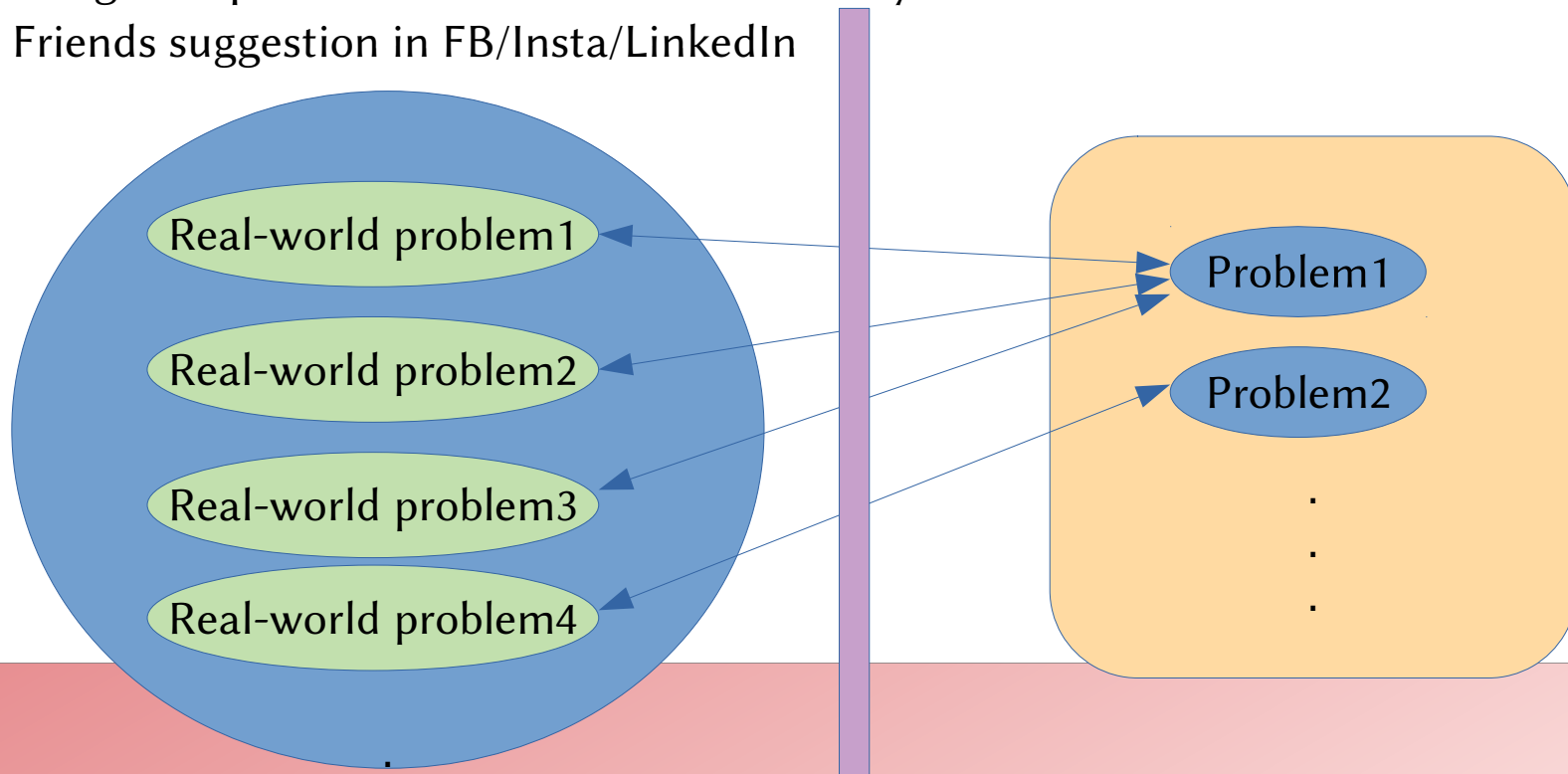


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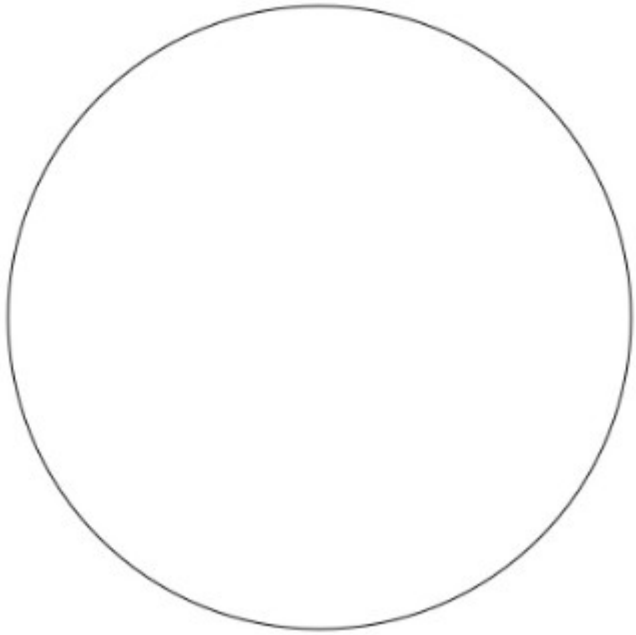
Abstraction

Researcher

- Google Map: Route from home to College
- Internet: Routing protocol
- Google Map: Show me the restaurants nearby
- Friends suggestion in FB/Insta/LinkedIn
- Shortest Path Problem
- Single Source Shortest Problem

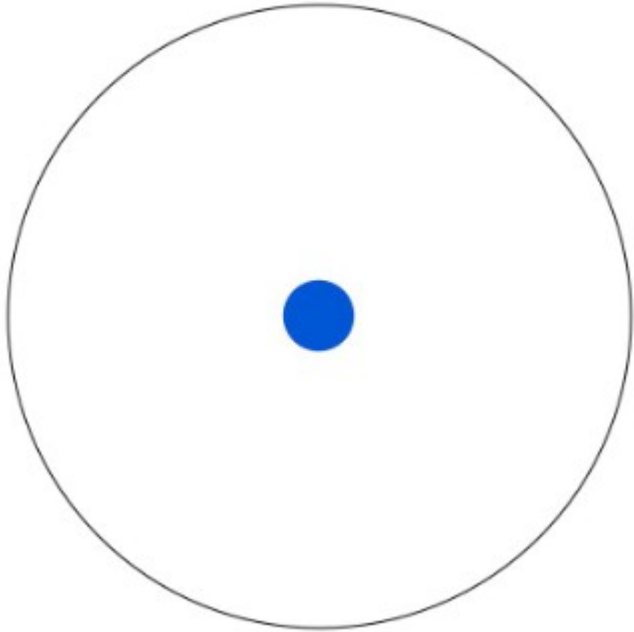


Research: 10000-feet view



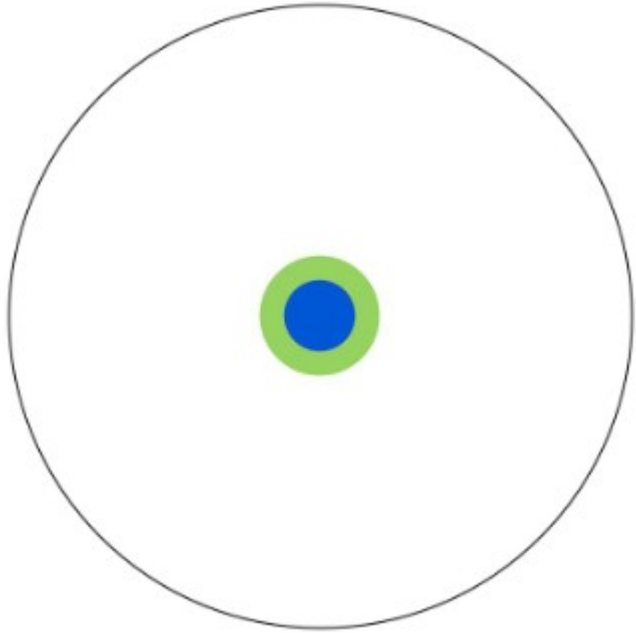
Prof. Matt Might's blog <https://matt.might.net/articles/phd-school-in-pictures/>

Research: 10000-feet view



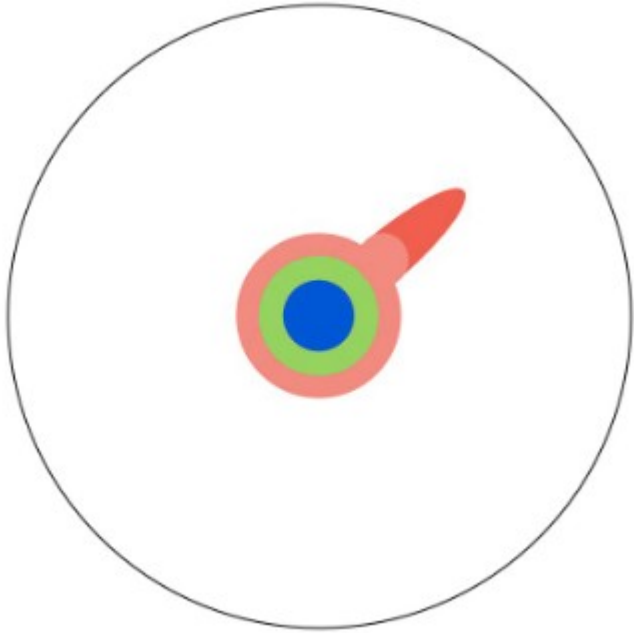
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Research: 10000-foot view



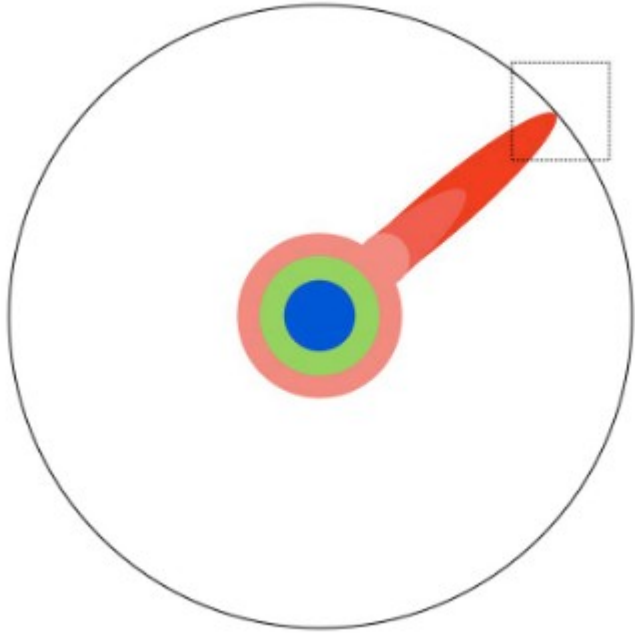
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Research: 10000-foot view



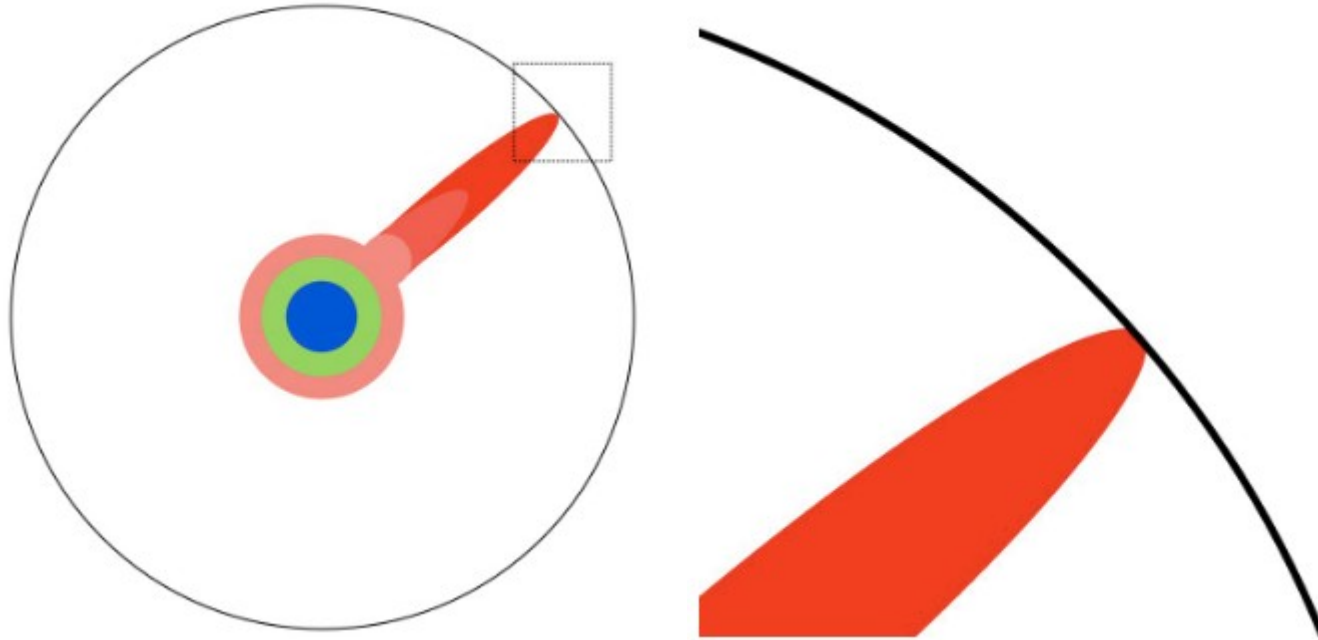
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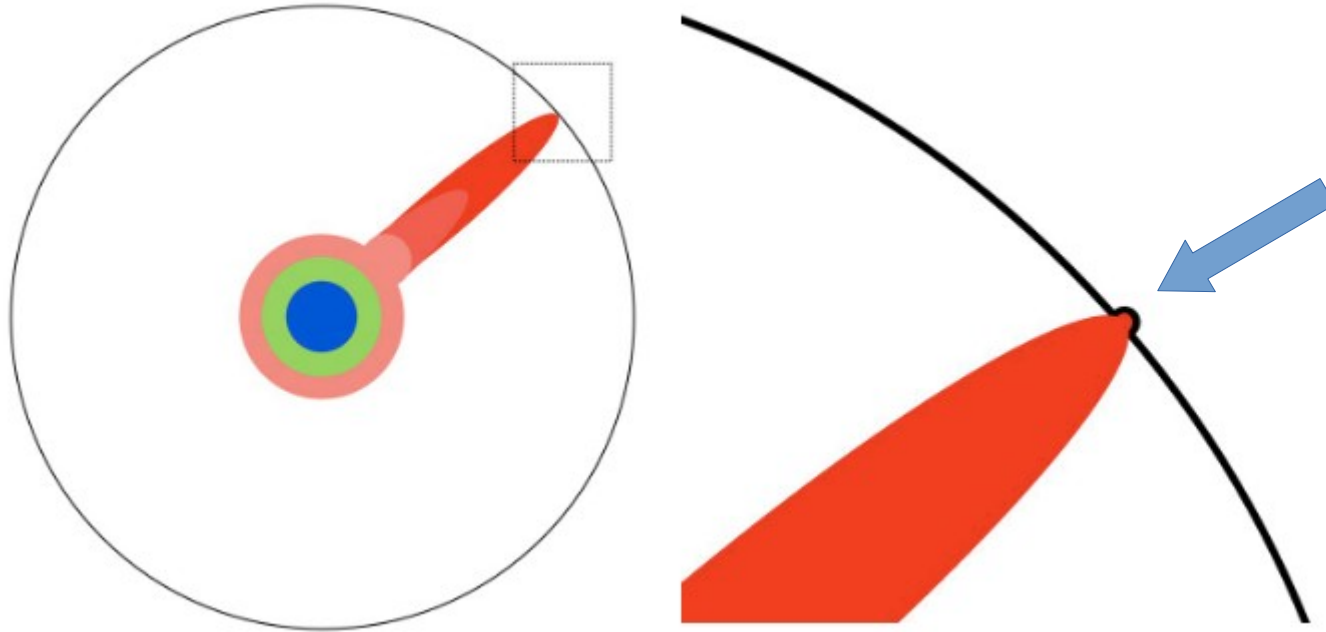
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Research: 10000-foot view



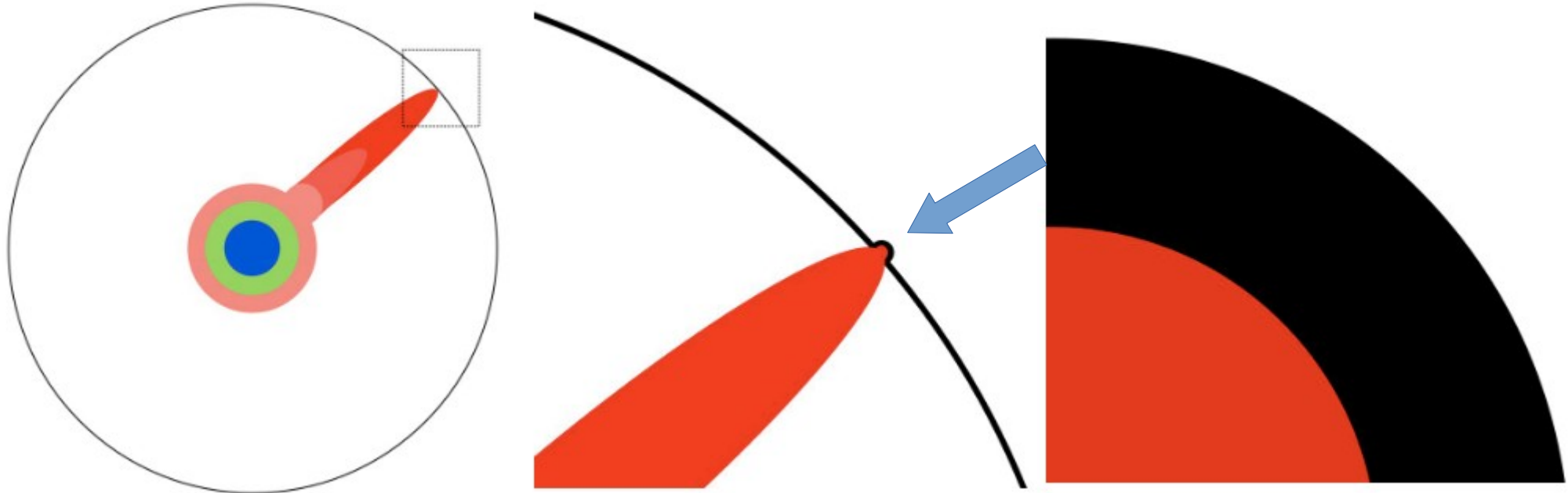
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Research: 10000-foot view



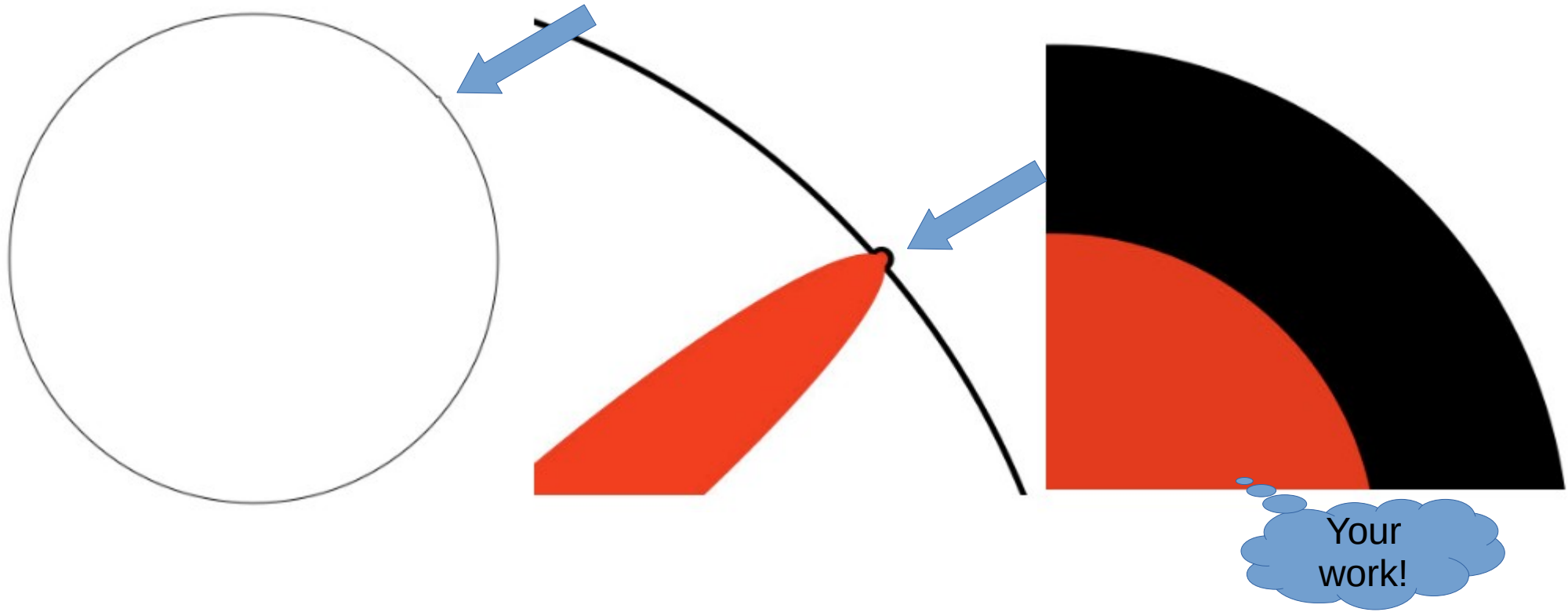
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How to choose research area/problem

- Most frequently asked question!



Courtesy: Tinkerbell [2008] Disney Animation Movie

How to choose research area/problem

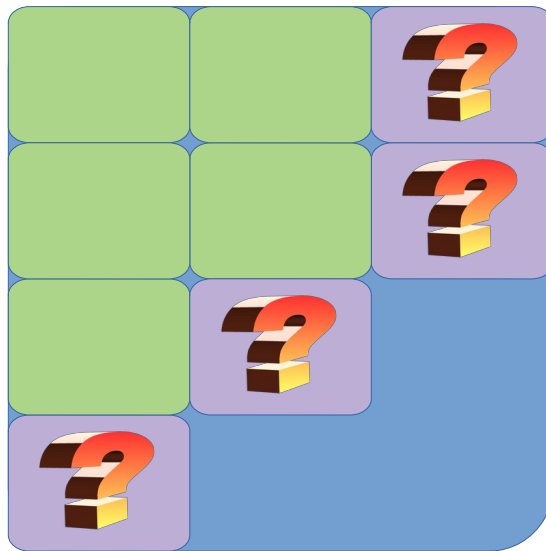
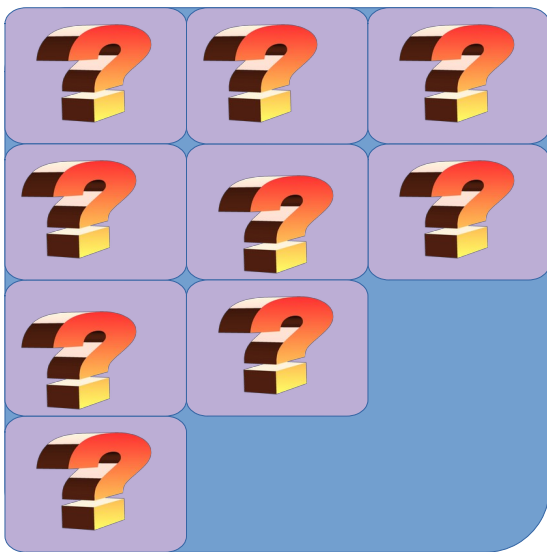
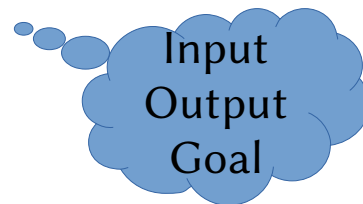
- Through Courses
- Attend research seminars/talks
- Attend conferences talks
- Read Survey papers

Useful Websites

- [Google Scholar](#)
- [DBLP](#)

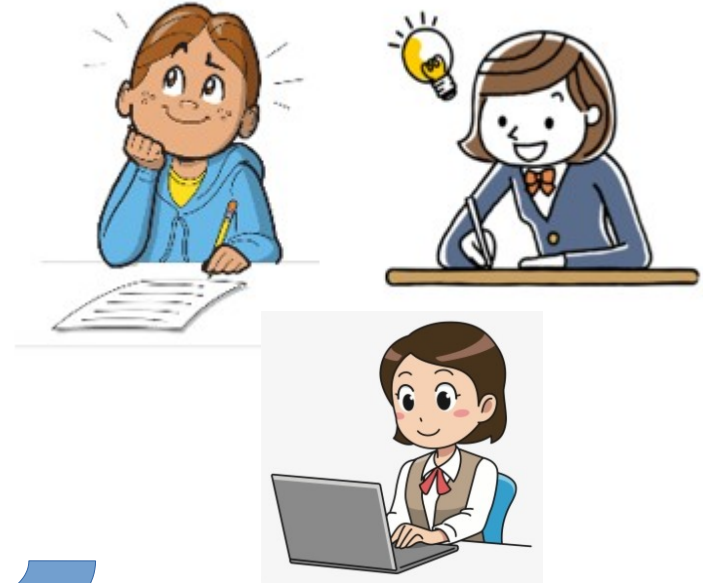
Doing research – Ask and answer question

- Play with small examples --> properties
- Ask a lot of small questions
- Answer some of those
- There is a bigger question //which no one knows the answer yet!

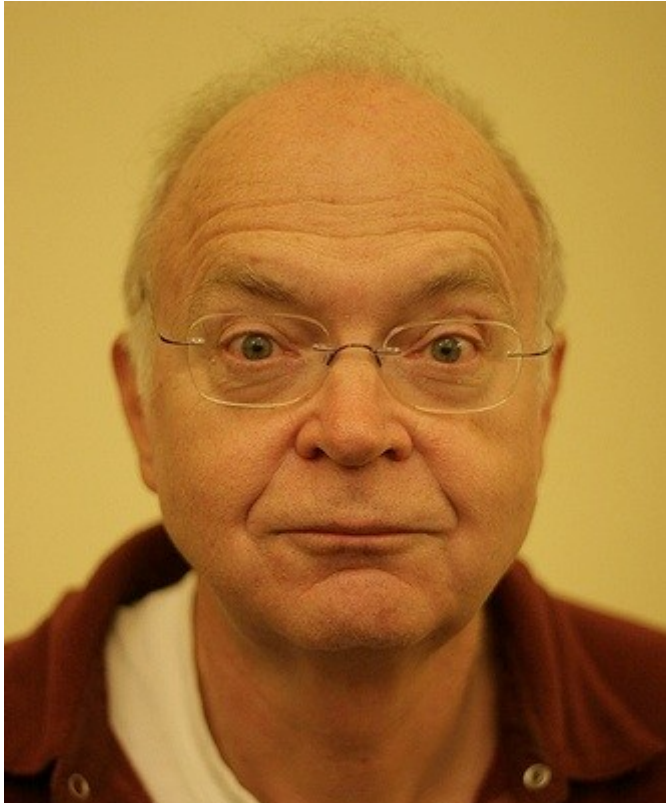


Discussion and Ideas

Ideas are
important



Writing up Research

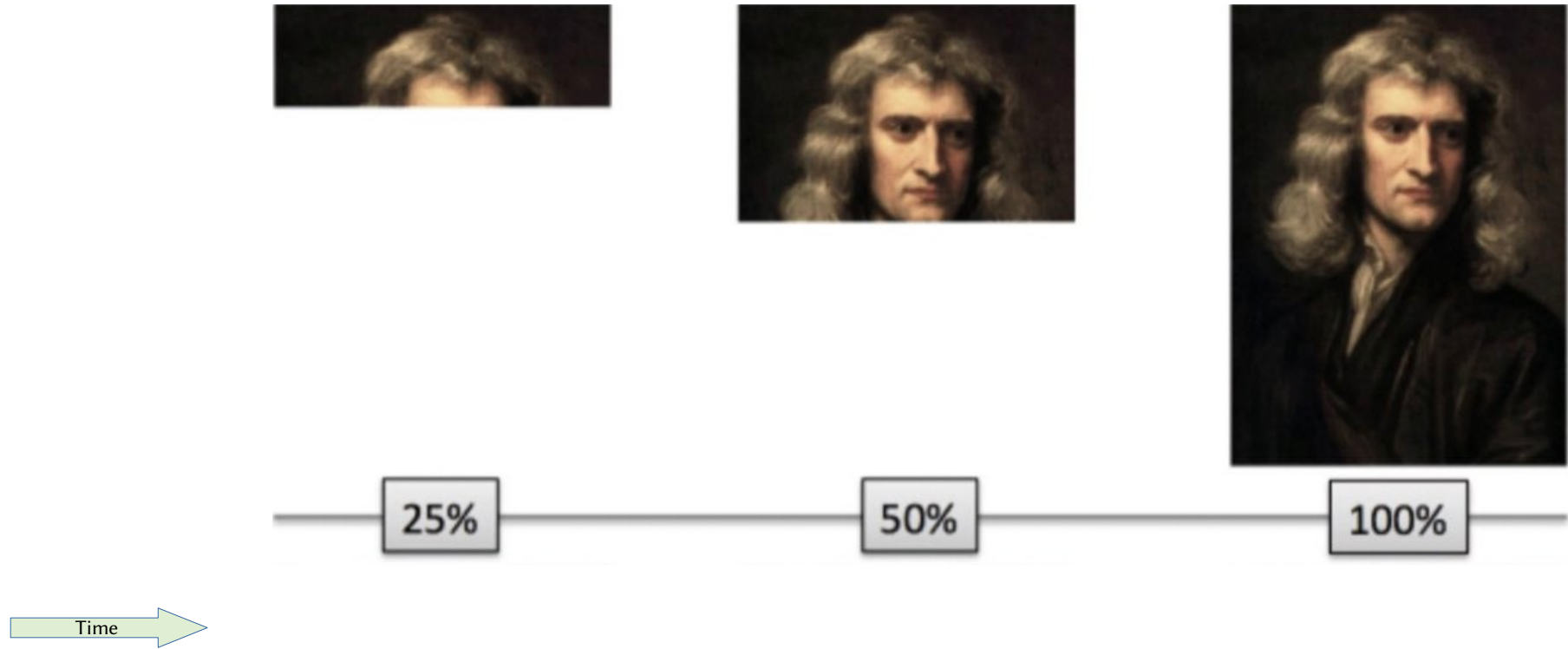


“Of equal importance to solving a problem is the communication of that solution to others.”
- Donald Knuth

Writing – Ways to write



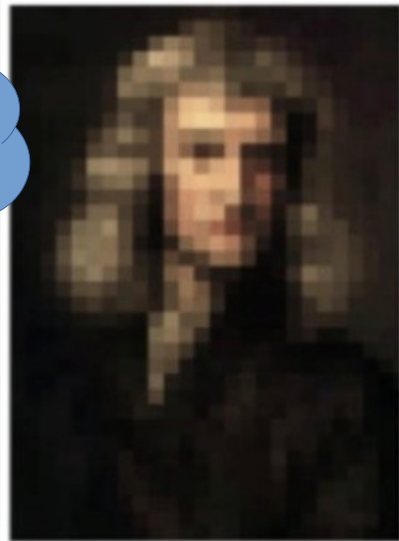
Writing – Method 1



Source: https://twitter.com/david_perell/status/1411852869804904450

Writing – Method 2

This method works for majority of people



25%



50%



100%

Time →

Source: https://twitter.com/david_perell/status/1411852869804904450

GPU - History

- GPU for Graphics
- CUDA v1 released 2006
- Seminal paper in GPU Research
- Donated GPUs for educational Insti.
- Researchers @ IIIT Hyderabad.
- GPU for general purpose computing → GPGPU



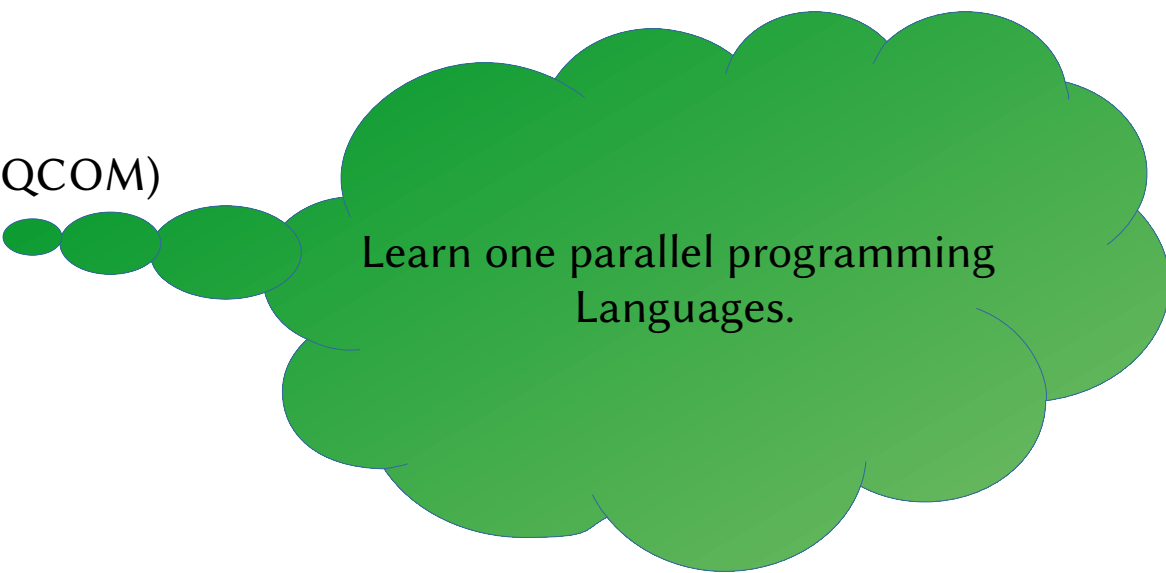
Parallel Programming languages/API

GPUs

- CUDA → Nvidia
- HIP → AMD
- SYCL → Intel
- OpenCL → All GPUs (Nvidia/AMD/Qualcomm)

CPUs

- OpenMP
- C++
- MPI



Learn one parallel programming Languages.

Current Philosophy

... take a **fresh look** at some of the classic graph algorithms and devise **faster** and more parallel GPU and CPU implementations.

- Fallin et al.

A High-Performance MST Implementation for GPUs

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ABSTRACT

Finding a minimum spanning tree (MST) is a fundamental graph algorithm with applications in many fields. This paper presents ECL-MST, a fast MST implementation designed specifically for GPUs. ECL-MST is based on a parallelization approach that unifies Kruskal's and Borůvka's algorithm and incorporates new and existing optimizations from the literature, including implicit path compression and edge-centric operation. On two test systems, it outperforms leading GPU and CPU codes from the literature on all of our 17 input graphs from various domains. On a Titan V GPU, ECL-MST is, on average, 4.6 times faster than the next fastest code, and on an RTX 3080 Ti GPU, it is 4.5 times faster. On both systems, ECL-MST running on the GPU is roughly 30 times faster than the fastest parallel CPU code.

CCS CONCEPTS

• Computing methodologies → Massively parallel algorithms.

KEYWORDS

Minimum spanning tree, minimum spanning forest, parallelism,

lines. In this example, the cheapest distribution grid that allows everyone to deliver or receive electricity is the MST shown.

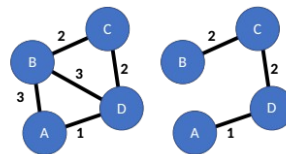


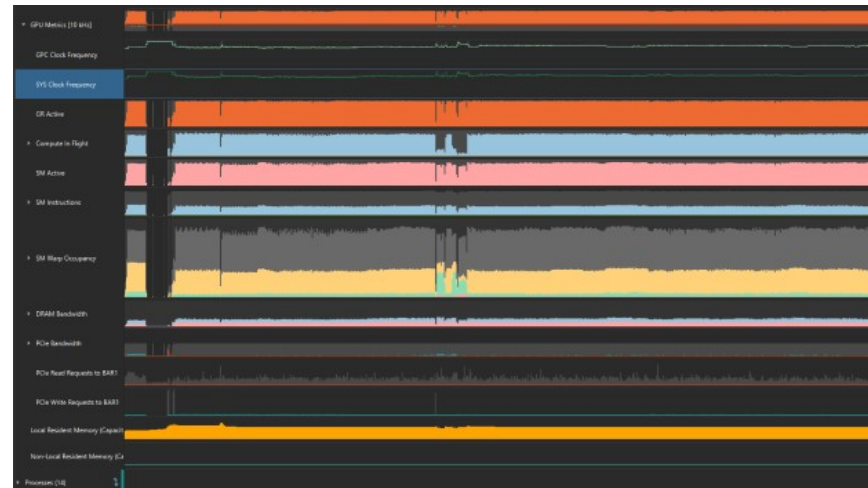
Figure 1: Example of a weighted graph on the left and the resulting MST on the right

Computing an MST (or MSF¹) is a fundamental graph algorithm with applications in many fields. For instance, it is a key building block in network analysis [12], chip design [1], eye tracking [17], route planning [13], and medical diagnostics like tumor recognition [4]. Since some of these applications repeatedly generate an MST, increasing the performance of this step is important and has the potential to speed up lifesaving computations.

SC'23

GPU Research

- GPU Programming course
 - Turning point
- People who are good at programming
- A decent laptop with NVD/AMD GPU
- NPTEL Course also available
- Initial setup is required.
- CUDA Programs
- Optimize for execution times.
- Use debuggers, profiling tools.



- People who understand GPU Arch
 - Are in high demand
 - Are widely respected.

AI – In Industry

- Major Players in semi-conductor
 - Nvidia
 - AMD
 - Google (TPU)
 - Qualcomm
- Several Accelerator startups in Bangalore
- Large models → smaller suitable models
- On device inference
 - Mobiles
 - Wearables
 - Humanoid robots

AI – Current trends

- Product Companies embrace AI
- People stopped coding line by line
- Large amounts of data
- Cleansing, scripting and reasoning is made faster
- Agents everywhere
- Domain knowledge is important
- People with mutple skills is respected
 - GPU+Compilers
 - Compilers+AI



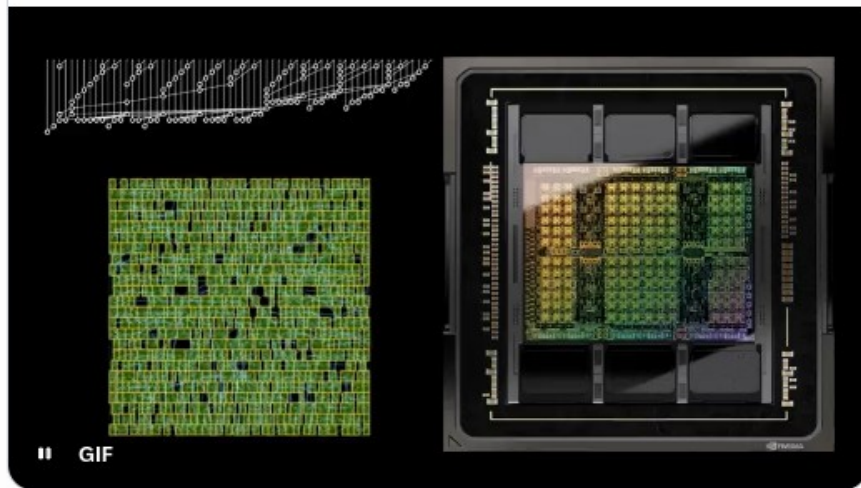
Bryce, the CUDA Colonel @blelbach · Jul 9, 2022

The GPUs are designing **themselves** now.



Rajarshi Roy @rjrshr · Jul 8, 2022

Arithmetic circuits were once the craft of human experts, and are now designed by AI in NVIDIA GPUs. H100 chips have nearly 13,000 AI designed circuits! How is this possible? [Blog developer.nvidia.com/blog/designing...](https://blog.developer.nvidia.com/blog/designing...) + a thread 📖💡



AI – General Guidelines for students

- Use it for learning new concepts like a tutor
- If program is not compiling or not giving expected output
 - Debug and get your hands dirt
 - Learn to use debuggers and profiling tools
- **One should not be dependant “too much” on AI.**
- Create a knowledge Wiki (or Second brain) of your notes
- Keep updating them for every courses for all semesters
- Use for revision for mid/final semester exams.
- I would strongly advice **against** using AI for programming assignments.
- Do hobby projects. You may use Agent to build, test, debug.
- Attend open house event in IITs and other similar events
- Attend workshops/symposiums/conferences from other colleges.

Key takeaways

- Cracking GATE and Getting into IITs is not that hard.
- Curious minded students do better than top rankers.
- Coming up with new ideas in research is important
- But, Writing up and presenting your research is even more important
- Use AI for learning concepts and build hobby projects. NOT for programming assignments
- Learn to use AI tools/agents but do NOT outsource the thinking/decision

References

- Donald Knuth, ACM Turing awardee [Post1](#) [Post2](#) [Post3](#)
- Manas Thakur, Assistant Professor, IIT Bombay. [Post1](#)
- Biswabandan Panda, Assistant Professor, IIT Bombay. [Post1](#) [Post2](#) [Post3](#)
- Thanks to my advisors, professors, students and friends at IIT Madras for the wisdom.
- Thanks to Google Images and twitter for pictures. :-)

Questions?



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